

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCPE506A - ADVANCED ENVIRONMENTAL ENGG.

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. What is the primary function of a waste stabilization pond (WSP)?
 - A) To decompose organic waste by mechanical means
 - B) To use microorganisms and sunlight to treat wastewater
 - C) To chemically neutralize hazardous waste
 - D) To provide a habitat for aquatic life
 2. In the Streeter-Phelps equation, what does the term 'BOD' stand for?
 - A) Biochemical Oxygen Demand
 - B) Biological Organic Degradation
 - C) Biodegradable Oxygen Demand
 - D) Biological Oxygen Depletion
 3. What happens when the oxygen demand of a water body exceeds its self-purification capacity?
 - A) The water body becomes clear and free of pollutants
 - B) The level of dissolved oxygen increases
 - C) The water body becomes polluted and fish and other aquatic life may die
 - D) The self-purification process speeds up
 4. Which of the following is NOT a common source of industrial wastewater pollution?
 - A) Textile mills
 - B) Power plants
 - C) Agriculture runoff
 - D) Chemical manufacturing plants
 5. Which of the following components is typically more prevalent in industrial wastewater compared to municipal wastewater?
 - A) Organic matter
 - B) Heavy metals
 - C) Nitrogen compounds
 - D) Pathogens

- 6 Oil flotation systems are most commonly used to treat wastewater that:
 - A) Has a high concentration of organic waste.
 - B) Contains large amounts of dissolved solids.
 - C) Contains free oil or grease.
 - D) Is highly acidic.
- 7 Which of the following methods is commonly used for phosphorus removal in wastewater treatment?
 - A) Nitrification.
 - B) Chemical precipitation.
 - C) Aeration.
 - D) Reverse osmosis.
- 8 Environmental sanitation refers to:
 - A) The process of cleaning public spaces.
 - B) The control of environmental factors that can adversely affect human health.
 - C) The management of solid waste in urban areas.
 - D) The installation of sewage systems only.
- 9 Proper solid waste management is crucial for maintaining:
 - A) Air quality only.
 - B) Public health and environmental sustainability.
 - C) Biodiversity preservation only.
 - D) Industrial waste recycling
- 10 What is the primary vector for malaria transmission?
 - A) Housefly
 - B) Mosquito (Anopheles species)
 - C) Rat
 - D) Tick
- 11 Where do houseflies commonly breed?
 - A) In dry, sunny places
 - B) In stagnant water
 - C) In decaying organic matter, such as garbage or manure
 - D) On trees and bushes
- 12 In rural areas, what is the primary source of drinking water?
 - A) Tap water connected to city pipelines
 - B) Bottled water purchased from stores
 - C) Wells, rivers, and handpumps
 - D) Water delivered by water tankers

Q2. Solve the following.

- A) Explain the principle & construction of stabilization pond with neat and clean diagram 6
- B) Discuss in detail about self purification of natural streams 6

Q3. Solve the following.

- A) Define industrial pollution. Write control measures to be undertaken to avoid industrial pollution 6
- B) What is neutralization? Explain any two methods of neutralization of industrial wastewater treatment 6

Q4. Solve Any Two of the following.

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| A) | Explain Nitrification and Denitrification process in industrial wastewater treatment | 6 |
| B) | What do you mean by CETP? Explain the need of CETP in industrial wastewater treatment | 6 |
| C) | Explain reverse osmosis process with neat and clean diagram | 6 |
- Q5. Solve Any Two of the following.
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| A) | What do you mean by environmental sanitation? Discuss in details about communicable diseases | 6 |
| B) | Explain in details about the diseases communicated by discharges of intestines | 6 |
| C) | Describe about the diseases communicated by nose and throat | 6 |
- Q6. Solve Any Two of the following.
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| A) | Explain Life cycle of Mosquito with neat diagram | 6 |
| B) | Discuss low cost excreta disposal systems | 6 |
| C) | Explain problems of water supply and sanitation aspects in rural areas | 6 |

*** End ***